An Interactive User-Centric Airline Recommendation System

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Introduction

Many travelers choose flights based on price and schedule, leaving uncertainty about service quality. Our project utilizes user feedback and an advanced algorithm to provide personalized airline suggestions.

Problem Definition

Our project introduces an intuitive web application that streamlines this process. Users receive **tailored airline suggestions** considering their preferences in areas like **food quality**, **seat comfort**, **entertainment**, **cabin service**, **ground service**, and **wifi connectivity**.

Data

The airline review data are combined with 2 large datasets (see detailed data links below) that were downloaded from Kaggle.com. In total, there are more than **195k data points** with **23 columns**, including **8 numeric attributes**, **13 categorical attributes**, and **2 timestamp attributes**. There are some missing values in the dataset, after detailed data cleaning there were **129,379 data points**. [**172 MB Size** on disk]

In addition, another data source with all city coordinates is obtained from https://www.partow.net/miscellaneous/airportdatabase/, including **44,691 major city names and their coordinates**. [**4.7 MB Size** on disk]

Airline Review Data Source 1: https://www.kaggle.com/datasets/efehandanisman/skytrax-airline-reviews/
Airline Review Data Source 2: https://www.kaggle.com/datasets/joelljungstrom/128k-airline-reviews

Interactive Visualization Approaches

What are they?

- Control Panel: Menu and User Preference Selection.
- Personalized Recommendation: The system suggests airlines based on the user's preference input.
- Map Visualization: Shows journey with pins and curved lines, along with departure, stopover, and arrival cities' names.
- Airline Exploration: Detailed service ratings and user comments.
- Review Submission: User feedback contributes to platform's data for dynamic database.

How do they work?

- Combination of HTML, CSS, JavaScript, and D3.js for front-end development.
- D3.js for dynamic, data-driven visualizations (world maps, gauge charts).
- Core JavaScript for interactive features (user preferences, review submissions, dynamic dropdown menus).
- CSS for styling, layout design, and responsive interface elements.
- Integration of these technologies ensures a user-friendly and visually appealing interface.

Why Effective?

- Enhances User Experience: Easy navigation and user-friendly interface.
- Personalization: Tailored airline suggestions based on user preferences.
- Intuitive Visualization: Journey mapping and gauge charts for clarity.
- Comprehensive Overview: Detailed airline features and reviews for informed decisions.

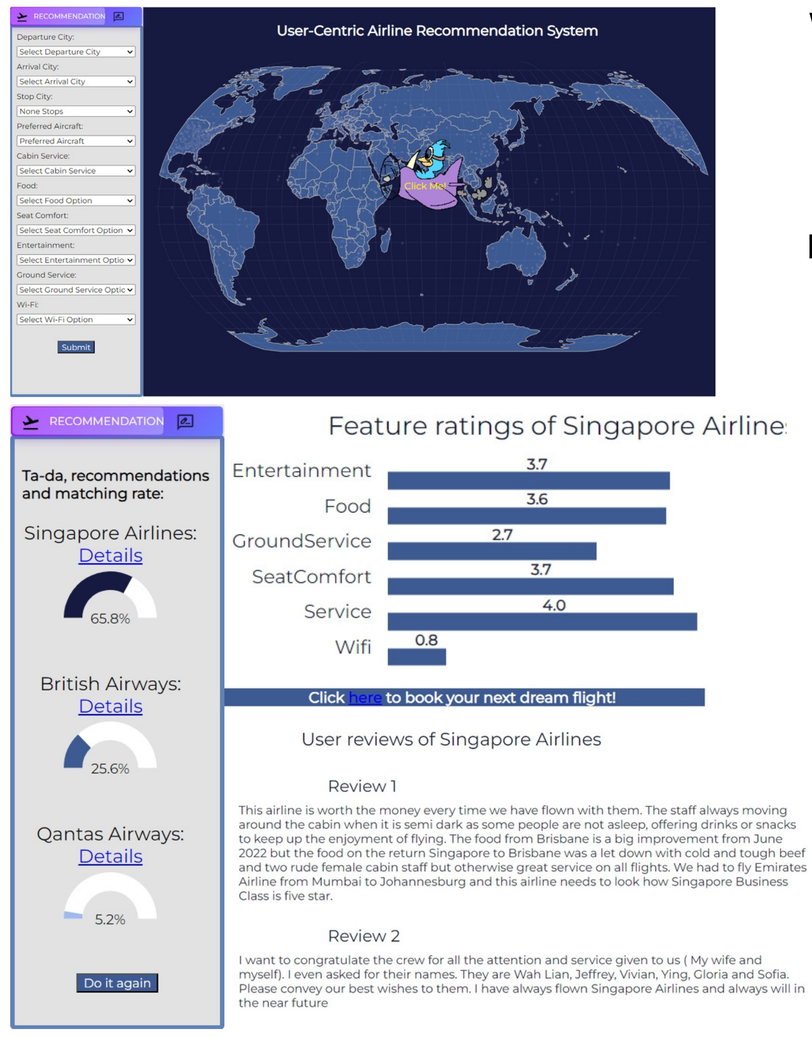

Figure 1. Interactive User Interface

Algorithm Approaches

What are they?

- User-Defined Metric Based Model.
- Data Modeling with Enhanced Normalization and Cleansing.
- Random Forest Classifier for Airline Suggestions.
- Integrated Validation and Parameter Optimization Techniques.

How do they work?

- Incorporates user-defined metrics (e.g., food, seat comfort).
- Utilizes mean imputation and one-hot encoding for data preparation.
- Random Forest Classifier evaluates top-3 airline suggestions.
- Out-of-Bag (OOB) accuracy for Random Forest model optimization.
- Synchronized integration with the user interface for real-time responses.

Why Effective?

- Tailored to User Satisfaction: Reflects crucial aspects of airline service.
- Precision and Proficiency: Random Forest Classifier ensures accuracy and timely response.
- Data Integrity: Comprehensive data cleansing for reliable modeling.
- User-Centric Design: The model adapts to individual user preferences in real-time.

Web Application Architecture

- Developed a WSGI (Web Server Gateway Interface) web application using Python Flask.
- Employed a backend service providing REST APIs that interact with SQLite database and recommendation model, transferring data between the frontend and the backend.

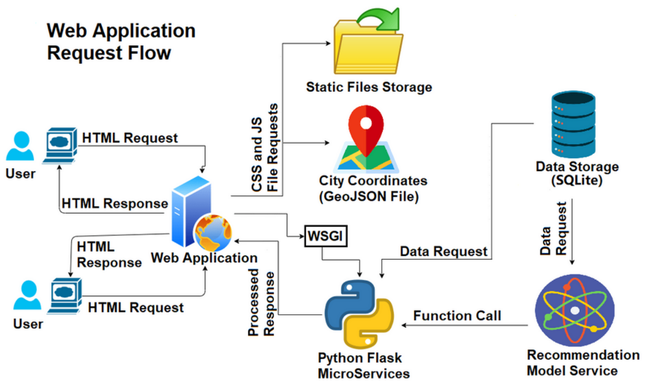

Figure 2. Web Application Architecture

What is new in our visualization and algorithm approaches?

- **Dynamic Geovisualization:** Real-time, interactive map feedback based on user-selected journey points (departure, connecting, and arrival cities).
- **Intuitive Gauge Charts for Airline Compatibility**: Simplifies decision-making by visually representing how well each airline matches user preferences.
- **Comprehensive Airline Exploration**: Detailed insights into airline services, user reviews, and direct links to official websites for well-informed choices.
- **Foundation for Future Enhancements**: Incorporates user reviews for database enrichment and model performance enhancement. Plans for algorithmic evolution to improve scalability and reduce training times.

Algorithm Evaluation

Evaluation Methods

- Multiple algorithms (RF, KNN, LR, RBF LR) were tested using various configurations with different data segments.
- Used 10-Cross Validation (10-CV) and Out-of-Bag (OOB) estimation for accuracy assessment (as shown in Table 1).

Evaluation Results

- The Random Forest (RF) model displayed consistently high accuracy.
- RF model's OOB accuracy indicated strong predictive strength.
- RF model proficiently included the correct label in its top 3 results, as highlighted in Table 2.

Method Comparison

- Demonstrated RF model's reliability and robustness for airline recommendation.
- Logistic Regression is identified as a potential solution for efficient processing in future expansions for project scaling.

No.	Model	Optimal HP	10-CV	Test
1	RF	t. num=29	0.8366 (OOB)	0.8769
	KNN	k=1	0.7717	0.6769
	LR	c=100	0.8234	0.8769
	RBF LR	g=0.1, c=100	0.6965	0.6923
2	RF	t. num=18	0.5478 (OOB)	0.6500
	KNN	k=17	0.5887	0.6250
	LR	c=50	0.5887	0.6500
	RBF LR	g=0.1, c=10	0.5650	0.6000
3	RF	t. num=23	0.5357 (OOB)	0.5714
	KNN	k=1	0.5214	0.5714
	LR	c=1	0.4571	0.5357
	RBF LR	g=0.1, c=100	0.5071	0.6429
4	RF	t. num=14	0.6535 (OOB)	0.8125
	KNN	k=1	0.6563	0.6563
	LR	c=1000	0.4976	0.4688
	RBF LR	g=0.1, c=1000	0.5713	0.5938

Table 1: Model Accuracy Comparison

Experiment No.	1	2	3	4
% correctly labeled	0.9846	0.8750	0.7500	0.9688

Table 2: Percentage of Correct Labels Included in Top-3 Recommendations by Random Forest

Visualization Evaluation

User Survey

- Diverse participants for user experience feedback.
- Surveyed aspects: ease of understanding, ease of use, visual appeal, airline recommendation, likeliness to use in the future, and likeliness to recommend to others.
- Rating scale: 1 (lowest) to 5 (highest).

Survey Results

- 53 participants completed the survey.
- Ratings of 4 and 5 were considered approvals of the web application.
- 96.3% found it easy to understand.
- 96.2% found it easy to use.
- 88.7% approved of the visual appeal.
- 90.6% were satisfied with airline recommendations.
- 92.4% likely to use in the future.
- 88.7% would likely recommend it to others.

User Textual Comments

- Approved of the uniqueness and innovation of this app.
- Suggested further improvement on visualization in the future.

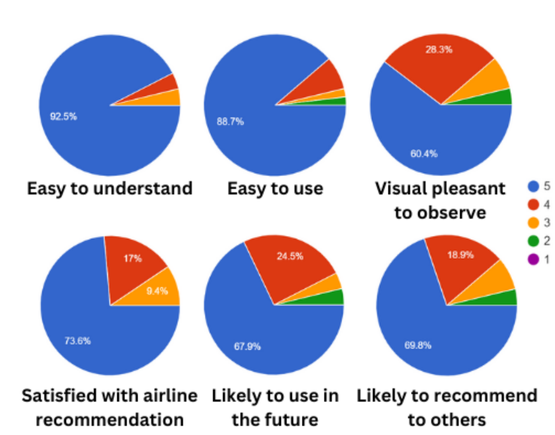

Figure 3. Visualization Evaluation Results